9.Index of First Occurrence in a String

AIM

To write a Python program to find the index of the first occurrence of a given substring (needle) in a main string (haystack) using the Brute Force String Matching algorithm.

PROCEDURE

1. Define a function to find the first occurrence of a substring in a given string.
2. Calculate the lengths of the main string and the substring.
3. Traverse the main string using a `for` loop.
4. Compare each character of the substring with the corresponding characters of the main string.
5. If all characters match, return the starting index.
6. If the substring is not found, return `-1`.
7. Display the first occurrence index.

main.py | Visualize | Run ▶ | Output

```python
def strStr(haystack, needle):
    n = len(haystack)
    m = len(needle)
    for i in range(n - m + 1):
        j = 0
        while j < m and haystack[i + j] == needle[j]:
            j += 1
        if j == m:
            return i
    return -1
haystack = "sadbutsad"
needle = "sad"
result = strStr(haystack, needle)
print("First Occurrence Index:", result)
```

```
First Occurrence Index: 0
```

RESULT

Thus, the Python program successfully finds the **index of the first occurrence of a substring** in the given string using the Brute Force String Matching algorithm.